

Lab 6: Differential Amplifier

Prep manual — what you are graded on, what to measure, and how to measure it

Sources: H&H *Learning the Art of Electronics* Lab 5L (§5L.1–5L.1.5), PHYS485/685 Lab 6 supplement, Lab 6 checklist

0 The whole lab on one page

Scope of today: only §5L.1, 5L.1.1, 5L.1.4, 5L.1.5. **Skip** 5L.1.3 (floating generator) and everything from 5L.1.6 on (homemade op-amp).

How you are graded. Your in-class grade is mainly the checklist. Each item needs an instructor signature; unlimited retries, no penalty. **Photograph the checklist and upload to ELMS at the end of every working session.**

Checklist item	What the TA wants to see	Numbers to have written down
1. 5L.1, 5L.1.1	Predicted gains on paper; CA3096 circuit wired (inputs A, B <i>not</i> connected yet)	G_{diff} , G_{CM} predicted (10k tail)
2. Transformer check (supplement) + 5L.1.4	Fig. 5L.7 input network built, <i>not</i> attached to amp; photos showing A/B behave correctly	Transformer diff. gain $(v_A - v_B)/v_{\text{pri}}$; transformer CM gain v_A/v_{CM} , v_B/v_{CM}
3. 5L.1.5, 10k Ω tail	Amp running, clean output	Measured G_{CM} , G_{diff} (+ compare to prediction)
4. 5L.1.5, JFET tail (E-152)	Tail resistor replaced by current-regulating diode	Measured G_{CM} , G_{diff} ; CMRR; written observations

Equipment. Keysight DSOX1102G (2 channels + built-in WaveGen on the front Gen Out BNC), external bench function generator, $\pm 15\text{ V}$ triple supply, DMM, breadboard, 6.3 V center-tapped transformer, CA3096 array, E-152 current-regulating diode.

Who does what:

- **External function generator** $\rightarrow$ transformer *primary* = **differential** signal. Its ground goes *only* to the primary (no ground loops).
- **Scope WaveGen** $\rightarrow 100\ \Omega \rightarrow$ transformer *center tap* = **common-mode** signal.
- Only 2 scope channels: you will keep moving probes. Plan: **CH1 = input (pin 1 / A)**, **CH2 = whatever you are checking** (B, or output pin 6).

1 Pre-lab: predictions (needed for sign-off #1)

1.1 DC operating point (do this first — everything else depends on it)

Inputs sit at $\approx 0\text{ V}$ DC (through the transformer winding), so

$$V_{\text{tail}} \approx -0.6\text{ V}, \quad I_{\text{tail}} = \frac{-0.6 - (-15)}{10\text{ k}\Omega} \approx 1.44\text{ mA}, \quad I_C \approx 0.72\text{ mA each}, \quad r_e = \frac{25\text{ mV}}{I_C} \approx 35\ \Omega,$$

$$V_{\text{out,DC}} = 15 - I_C R_C \approx 15 - 7.2 \approx 7.8\text{ V}.$$

Why it matters: r_e sets the gain (the book says “don’t neglect r_e ”), and $V_{\text{out}} \approx \text{mid-supply}$ tells you the output has room to swing both ways.

1.2 Differential gain — half-circuit argument

Drive $v_A = +v_d/2$, $v_B = -v_d/2$. By symmetry one emitter current goes up exactly as much as the other goes down, so the tail node *doesn’t move*: it is an AC ground. Each side is then a CE stage with emitter impedance

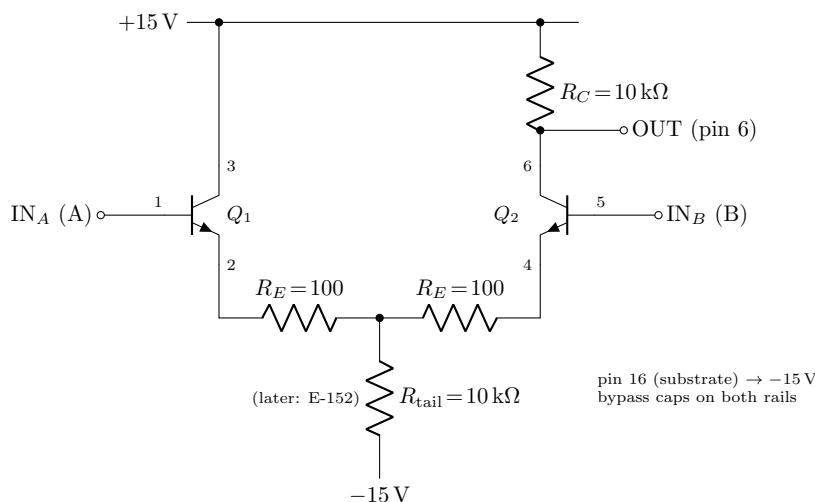


Fig. 5L.1 with CA3096 pin numbers. Output is single-ended at pin 6. Q_1 is the left NPN (B1, E2, C3), Q_2 the next one (E4, B5, C6).

$R_E + r_e$, and Q_2 sees only $-v_d/2$:

$$G_{\text{diff}} = \frac{v_{\text{out}}}{v_A - v_B} = \frac{R_C}{2(R_E + r_e)} \approx \frac{10\text{ k}}{2(135)} \approx 37$$

Output is *in phase with A* (A up $\Rightarrow$ tail up $\Rightarrow Q_2$'s V_{BE} down $\Rightarrow I_{C2}$ down $\Rightarrow V_{\text{out}}$ up). The supplement's "factor of 40" is this number. Single-ended output is why you get the factor of 2.

1.3 Common-mode gain — split the tail

Drive $v_A = v_B = v_{\text{CM}}$. Now both emitter currents move together, so the tail node *does* move. Split R_{tail} into two $2R_{\text{tail}}$ in parallel, one per side; each side is a CE stage with emitter impedance $R_E + r_e + 2R_{\text{tail}}$:

$$G_{\text{CM}} = \frac{v_{\text{out}}}{v_{\text{CM}}} = -\frac{R_C}{2R_{\text{tail}} + R_E + r_e} \approx -\frac{10\text{ k}}{20\text{ k}} \approx -0.5$$

Inverted: output is 180° from the (in-phase) inputs. (The supplement warns Fig. 5L.8 is misleading on this.)

1.4 CMRR and the current-source upgrade

$$\text{CMRR} = \left| \frac{G_{\text{diff}}}{G_{\text{CM}}} \right| \approx \frac{R_{\text{tail}}}{R_E + r_e} \approx 74 (\approx 37\text{ dB}), \quad \text{CMRR}_{\text{dB}} = 20 \log_{10} \text{CMRR}.$$

Replace R_{tail} with a current source of dynamic impedance Z_{cs} ($\sim \text{M}\Omega$ for a JFET CRD — check the current-regulating-diode datasheet on ELMS):

$$G_{\text{CM}} \approx -\frac{R_C}{2Z_{\text{cs}}} \sim -\frac{10\text{ k}}{2\text{ M}} = -0.005, \quad G_{\text{diff}} \text{ essentially unchanged } (I_{\text{tail}} = 1.5\text{ mA} \Rightarrow r_e \approx 33\Omega, G_{\text{diff}} \approx 37.6).$$

The *point*: G_{diff} depends only on $R_E + r_e$ (tail is AC ground anyway), while G_{CM} depends on the tail impedance. Make the tail stiff and only G_{CM} drops. Expect your measured G_{CM} to be bigger than 0.005 — transistor/resistor mismatch and capacitive feedthrough (worse at high f) leak differential gain in.

Checklist sign-off #1, part a: Write on your sheet: $V_{\text{out,DC}}$, I_{tail} , r_e , $G_{\text{diff}} \approx 37$, $G_{\text{CM}} \approx -0.5$, and the JFET-tail G_{CM} estimate.

2 5L.1.1 — Wire the CA3096 (sign-off #1, part b)

No measurements yet — just build it right.

1. Look up the CA3096 datasheet pinout (Fig. 5L.2). Find pin 1 (dot/notch). **IC straddles the center trench** — otherwise pins on opposite sides share a breadboard row and short together (that’s the “why?”).
2. **Pin 16 (substrate)** $\rightarrow -15\text{ V}$. The substrate forms parasitic diodes to every transistor; tying it to the most negative point keeps them reverse-biased.
3. Power buses: one pair = $+15/\text{GND}$, other pair = $-15/\text{GND}$. Put a bypass cap from each rail to ground near the IC (e.g. $0.1\text{ }\mu\text{F}$ ceramic, optionally $\sim 10\text{ }\mu\text{F}$ electrolytic — *watch polarity on the -15 rail: $+$ lead to GND*). This circuit likes to oscillate; caps and short leads help.
4. Wire Fig. 5L.1 using the pin numbers above. Leave IN_A (pin 1) and IN_B (pin 5) **unconnected** (they will *not* go straight to a function generator).
5. Set supply to $\pm 15\text{ V}$ *before* connecting, then power up.

Quick DC sanity check (DMM, 30 s, saves an hour): ground pins 1 and 5 temporarily with jumpers, then measure

Node	Expect	If not
Tail node (between $100\text{ }\Omega$ s)	$\approx -0.6\text{ V}$	wrong pins / transistor not conducting
Pin 6 (output)	$\approx 7.8\text{ V}$	$\approx 15\text{ V}$: Q_2 off; ≈ 0 : wiring short
Across each $100\text{ }\Omega$	$\approx 70\text{ mV}$ ($\Rightarrow 0.7\text{ mA}$)	unequal $\Rightarrow$ mismatch/miswire

Remove the grounding jumpers afterward.

Checklist sign-off #1, part b: Circuit wired on the CA3096, powered, inputs A/B open.

3 Input network + transformer check (sign-off #2)

Do NOT attach the transformer to the amplifier in this section. You are characterizing the transformer by itself. Use 5L.1.4 (Fig. 5L.7); do **not** do 5L.1.3. Never plug the transformer into the wall.

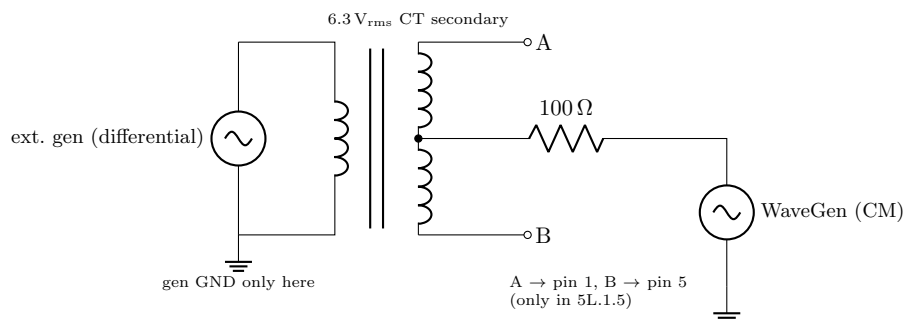


Fig. 5L.7: external generator $\rightarrow$ primary (difference signal); scope WaveGen $\rightarrow 100\text{ }\Omega \rightarrow$ center tap (common mode).

Setup. Both generators: 1 kHz sine, about the same amplitude (start $\sim 2\text{--}3\text{ V}$). Scope probe grounds $\rightarrow$ circuit ground.

3.1 Test A — differential only: A and B should be 180° apart

1. Disconnect the WaveGen/ $100\text{ }\Omega$ from the center tap and **short the center tap to ground** (for this transformer-only test the supplement also suggests tying it to the external generator’s ground — that makes the picture cleanest).

2. Ext. gen: large-ish signal (a few V). CH1 on A, CH2 on B.
3. Expect two equal sines, 180° out of phase. Read phase: [Meas] $\rightarrow$ add *Phase* (source 1 vs 2). Read amplitudes: [Meas] $\rightarrow$ *Peak-Peak* on each channel.
4. Turn on [Math] $\rightarrow$ operator $1 - 2$. This trace is $v_A - v_B$. Measure its pk-pk.
5. Move CH2 to the primary (hot side, ground clip to gen ground) to get v_{pri} . Return CH2 to B.

Transformer differential gain $= \frac{(v_A - v_B)_{\text{pp}}}{v_{\text{pri,pp}}} \approx \frac{1}{19}$ (6.3 V out of 120 V). Each half: $v_A/v_{\text{pri}} \approx 1/38$. Unloaded and driven at low voltage it may read slightly higher; either way, call it “ $\times \frac{1}{20}$ ”. **Photo this screen.**

3.2 Test B — common mode only: A and B in phase, equal

1. Remove the center-tap short; reconnect WaveGen $\rightarrow 100\Omega \rightarrow$ center tap.
2. Ext. gen amplitude to minimum (or unclip it and short the primary leads together).
3. CH1 on A, CH2 on B: two identical, in-phase sines; Math (1–2) should be $\approx$ flat.
4. Move one probe to the WaveGen side of the 100Ω for v_{CM} .

Transformer CM gain $= v_A/v_{\text{CM}} \approx v_B/v_{\text{CM}} \approx 1$ (no load, so the 100Ω drops nothing). **Photo this screen.**

3.3 Test C — independent control

Turn both generators on. Turn only the ext. gen knob: A and B move *oppositely*. Turn only the WaveGen amplitude: A and B move *together*. Tip: set them to different frequencies (e.g. diff 1 kHz, CM 100 Hz) to see both at once. **Photo.**

Checklist sign-off #2: Photos: (i) A, B 180° apart with Math $A - B$ showing; (ii) A, B in phase, Math ≈ 0 ; (iii) mixed. Numbers: transformer diff. gain and CM gain.

4 Scope + generator settings you will reuse

Task	DSOX1102G method
Probe setting	Channel key [1]/[2] $\rightarrow$ <i>Probe</i> $\rightarrow$ match the probe (10:1 for $10\times$ probes). Wrong setting = gain off by 10.
See a small wiggle on 7.8 V DC	Channel menu $\rightarrow$ <i>Coupling</i> : <i>AC</i> on the output channel, then turn V/div down.
Amplitudes	[Meas] $\rightarrow$ <i>Peak-Peak</i> (or <i>Amplitude</i>). Use pk-pk consistently on both channels; gains are ratios so the convention cancels.
Phase	[Meas] $\rightarrow$ <i>Phase</i> , source 1 and 2. $\approx 0^\circ$ or $\approx \pm 180^\circ$ is what you are checking.
Difference $v_A - v_B$	[Math] $\rightarrow 1 - 2$.
Noisy, tiny signal (JFET CM output)	[Acquire] $\rightarrow$ <i>Averaging</i> (16–64); channel menu $\rightarrow$ <i>BW Limit</i> on. Trigger on the <i>big</i> channel (CH1).
Freeze a drifting mixed display	[Run/Stop] or [Single].
WaveGen	[Wave Gen] $\rightarrow$ Sine, <i>Frequency</i> , <i>Amplitude</i> . Amplitude is peak-to-peak ; set output load to <i>High-Z</i> so the number matches what you get. Max $\approx 10\text{ V}_{\text{pp}}$ — that is the supplement’s “as large as 10 V”.
Save screenshot	[Save/Recall] $\rightarrow$ <i>Save</i> $\rightarrow$ PNG to USB stick (or phone photo of the screen).
Spot oscillation	Output looks fuzzy/thick? Drop timebase to $\mu\text{s}/\text{div}$. MHz hash = oscillation $\rightarrow$ shorten leads, check bypass caps, keep the input probe on pin 1.

5 5L.1.5 — Diff amp with 10 k Ω tail (sign-off #3)

Now connect A $\rightarrow$ pin 1, B $\rightarrow$ pin 5. Both generators 1 kHz, similar amplitude.

Probe the input at pin 1, not pin 5. Pin 5 is the base of the gain side (Q_2); a probe's capacitance there tends to make it oscillate. Since $v_A \approx \pm v_B$ by symmetry, pin 1 tells you everything.

5.1 Common-mode gain G_{CM}

1. Ext. gen amplitude to minimum (differential off).
2. WaveGen: 1 kHz, a few V (e.g. 4–6 V_{pp}).
3. CH1 $\rightarrow$ pin 1 (DC coupled), CH2 $\rightarrow$ pin 6 (AC coupled).
4. Measure $v_{in,pp}$ (CH1), $v_{out,pp}$ (CH2), and phase.

$$G_{CM} = -\frac{v_{out,pp}}{v_{in,pp}} \quad (\text{expect } \approx -0.5, \text{ phase } \approx 180^\circ).$$

If it reads too high: some differential signal is sneaking in. Short A to B with a wire (book: parallel the inputs) so the input is *forced* to be pure CM, re-measure, **then remove the wire**.

5.2 Differential gain G_{diff}

1. Supplement change: **disconnect the center tap from the WaveGen/100 Ω and connect it to ground** (instead of “cutting CM to a minimum”).
2. Ext. gen: 1 kHz. The transformer divides by ~ 20 and the amp multiplies by ~ 40 , so you need a few volts on the primary (the instructor used ≈ 2.6 V). Raise it until the output is a few volts and *clean* — no flattened tops (clipping).
3. CH1 $\rightarrow$ pin 1, CH2 $\rightarrow$ pin 6 (AC). Measure $v_{A,pp}$, $v_{out,pp}$, phase ($\approx 0^\circ$).
4. Because the CT is grounded, $v_B = -v_A$, so $v_A - v_B = 2v_A$:

$$G_{diff} = \frac{v_{out,pp}}{2v_{A,pp}} \quad (\text{expect } \approx 37).$$

Classic mistake: using v_{out}/v_A gives ≈ 75 — the “factor of two too high” the book warns about. If in doubt, move CH2 to B for a moment and use Math (1–2) to read $v_A - v_B$ directly. Cross-check: $v_A - v_B \approx v_{pri}/20$.

5.3 Mixed signals (observe, then write a few lines)

Reconnect the CT to WaveGen/100 Ω . Diff at 1 kHz, CM at a different frequency (e.g. 100 Hz) so you can tell them apart. Slowly dial CM from minimum up to several volts while watching pin 6. You should see the amplified 1 kHz signal riding on an inverted, $\sim \times 0.5$ copy of the CM signal — i.e. CM is rejected only $\sim 74\times$ relative to the difference, so a big CM signal is plainly visible. **Photo + 1–2 sentences.**

Checklist sign-off #3: Measured G_{CM} and G_{diff} with 10 k tail, compared to prediction.

6 5L.1.5 — JFET current-source tail (sign-off #4)

1. Power off. Replace the 10 k tail resistor with **one E-152** (1.5 mA; it replaces the book's two 0.75 mA 1N5294s in parallel).
2. **Orientation:** current flows from the tail node *down* to -15 V, so anode $\rightarrow$ tail node, cathode (band) $\rightarrow -15$ V. Confirm on the ELMS datasheet. Backwards, it conducts like an ordinary diode and the only thing limiting current is the two 100 Ω s — you will cook the array.
3. Power on, DC check: ≈ 75 mV across each 100 Ω (0.75 mA each), $V_{out} \approx 7.5$ V.

6.1 G_{CM} (JFET)

Same method as before, but the output is now tiny, so push the input up: WaveGen up to $\sim 10 V_{\text{pp}}$, ext. gen at minimum (or primary leads shorted together). CH2 on pin 6: AC coupling, 10–20 mV/div, averaging on, trigger on CH1.

$$G_{\text{CM}} = -\frac{v_{\text{out,pp}}}{v_{\text{in,pp}}} \quad (\text{ideal} \sim 0.005; \text{ expect somewhat more}).$$

Worth doing: repeat at 100 Hz and 10 kHz. CMRR is best at low f and degrades as f rises (capacitive coupling, §5L.1.11). One sentence on this makes a good report point.

6.2 G_{diff} (JFET)

Identical to Section 5.2 (CT to ground, ext. gen on, $G_{\text{diff}} = v_{\text{out}}/2v_A$). Expect $\approx$ unchanged (~ 37 – 38). That “unchanged” is the result: the tail only affects CM.

6.3 Explicit CMRR + swap test (write text)

- Compute $\text{CMRR} = |G_{\text{diff}}/G_{\text{CM}}|$ and in dB for both tails.
- Mixed signals again (diff + CM). Swap tail between 10 k and E-152 (power off each time) and turn the CM amplitude up/down. With the resistor the CM component is obvious; with the JFET the output barely notices it. Write a short paragraph on what you see.

IC removal: pull straight up, both sides at once, so pins don’t bend.

Checklist sign-off #4: Measured G_{CM} and G_{diff} with JFET tail; CMRR; written swap observations.

7 Data sheet (fill in at the bench)

Quantity	Predicted	$v_{\text{in,pp}}$	$v_{\text{out,pp}}$	Measured
DC: $V_{\text{tail}} / V_{\text{out}}$ (10 k)	−0.6 / 7.8 V	—	—	_____
Transformer diff. gain $(v_A - v_B)/v_{\text{pri}}$	$\approx 1/19$	_____	_____	_____
Transformer CM gain $v_A/v_{\text{CM}}, v_B/v_{\text{CM}}$	≈ 1	_____	_____	_____
G_{CM} , 10 k tail (phase _____)	−0.5	_____	_____	_____
G_{diff} , 10 k tail ($= v_{\text{out}}/2v_A$)	37	_____	_____	_____
CMRR, 10 k tail (ratio / dB)	74 / 37 dB	—	—	_____
DC: V_{out} (JFET)	7.5 V	—	—	_____
G_{CM} , JFET @ 1 kHz	~ 0.005	_____	_____	_____
G_{CM} , JFET @ 100 Hz / 10 kHz	—	_____	_____	_____
G_{diff} , JFET	37.6	_____	_____	_____
CMRR, JFET (ratio / dB)	~ 7500 / 77 dB	—	—	_____

Photos to have: checklist (each session); transformer A/B anti-phase + Math; A/B in phase; mixed signal at output with 10 k tail; mixed signal with JFET tail.

Text to write: (1) does each gain match prediction, and if not, why (mismatch, r_e estimate, probe loading, transformer ratio); (2) what happens as CM is dialed up, resistor vs. JFET; (3) CMRR vs. frequency.

Troubleshooting, fast

Symptom	Likely cause
Output stuck at $\sim 15\text{ V}$	Q_2 off: tail/emitter wiring, pin 16 not at -15 , wrong pins
Output fuzzy / thick trace	Oscillation: bypass caps, short wires, probe on pin 1 not pin 5
$G_{\text{diff}} \approx 75$	Forgot the factor of 2 ($v_A - v_B = 2v_A$)
G_{diff} off by 10	Probe attenuation setting doesn't match probe
G_{CM} way above 0.5	Differential leaking in: short A–B to confirm
Flat-topped output	Clipping: reduce ext. gen amplitude
Hum / weird baseline	Ground loop: ext. gen ground touching circuit ground
